TT-P-55B
AMENDMENT 2
September 21, 1972
SUPERSEDING
AMENDMENT 1
March 18, 1971

FEDERAL SPECIFICATION

PAINT, POLYVINYL ACETATE EMULSION, EXTERIOR

This amendment, which forms a part of Federal Specification TT-P-55B, dated April 27, 1965, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

Page 1

Under 2.1, Federal Specifications, delete all reference to TT-P-442.

Following 2.1, add:

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D476	- Titanium Dioxide Pigments.
D2088	- Determination of Low Concentrations of Lead in Paint.
D2206	- Determination of Low Quantities of Mercury in Paint.
E-2 SM10-19	- Methods of Emission Spectrochemical Analysis.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Page 2

3.2.2, lines 7 and 8, delete "TT-P-442, type III, grade A," and substitute "ASTM D476, type IV,".

Table I, under "Material", line 2, delete "TT-P-442, type I" and substitute "ASTM D476, type I"; lines 6 and 7, delete "TT-P-442, type III, grade A" and substitute "ASTM D476, type IV".

FSC 8010

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:14Z
Check the source to verify that this is the current version before use.

Table II, after "Mica, pounds per gallon of paint" insert in respective columns:

Characteristic	Requirement	
	Minimum	Maximum
"Lead, percent by weight of total solids	-	0.5
Mercury, percent by weight of paint	None	None "

Page 3

3.4.7, delete and substitute:

3.4.7 Fungus resistance. The paint shall not support the growth of fungus when tested in accordance with 4.3.9.

Page 4

Table IV, lines 8 and 9, under column titled "Test method", delete "4141 and 4142" and substitute "3021 and 3022"; at end of table add in applicable columns:

"Fungus resistance 6271 4.3.9"

4.3.4, delete and substitute:

4.3.4 Mica. Mica content of the paint shall be determined in accordance with either 4.3.4.1 or 4.3.4.2.

4.3.4.1 Precipitation method. Weigh out 0.5 g of the extracted pigment and mix well with 0.5 g ammonium chloride and 3 g calcium carbonate. Transfer to a platinum crucible, placing 0.5 g of calcium carbonate in the bottom of the crucible and a similar amount over the top of the mixture. Heat gently over a low flame for between 12 and 15 minutes to volatilize ammonium chloride. Then raise the temperature until the lower three-quarters of the crucible is dull red. Hold this temperature for about 1 hour. Cool, take up with 50 cc of water, and heat over a water bath adding water to replace that lost by evaporation. Break up any lumps with a small pestle. Decant the clear liquid through a filter paper and wash four times by decantation. Then transfer the residue to the filter. Wash until silver nitrate shows only a very faint turbidity. To the filtrate, add ammonium hydroxide and ammonium carbonate and heat to boiling. Filter and digest the precipitate with ammonium hydroxide and ammonium carbonate. Filter and allow the filtrate to collect with the previous filtrate, then evaporate to dryness in

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:14Z
Check the source to verify that this is the current version before use.

a platinum or porcelain dish. Remove ammonium salts by gentle ignition in moving flame. After the residue cools, dissolve it in a little water and add a few drops of ammonia and ammonium oxalate. Let stand overnight. Filter, evaporate to dryness, ignite gently, and allow to cool. Then moisten the residue with hydrochloric acid, evaporate again to dryness, ignite gently, and weigh. (The increase in weight is potassium chloride plus sodium chloride.) Dissolve in water. (Ignite and weigh any insoluble residue, deducting this from the mixed chlorides.) Dilute the solution, and add hydrochloroplatinic acid in amount equal to 0.3 ml more than would be necessary if all chlorides were present as sodium chloride. Evaporate to a syrupy consistency. Cool and treat with a few ml of 80 percent ethyl alcohol. Stir and decant through a weighed Gooch crucible. Treat again with 80 percent alcohol and decant. Repeat washings and decantation until the alcohol leaving the Gooch crucible is colorless and the precipitate appears golden yellow. Transfer the precipitate to the Gooch crucible, wash with 80 percent alcohol six times, and dry at plus 130° C. to constant weight as potassium chloroplatinate. Calculate as follows:

$$\frac{\text{Weight of potassium chloroplatinate} \times 0.9376 \times 100}{\text{Weight of sample}} = \text{percent of potassium oxide}$$

Percent of $K_2O \times 8.45$ = percent mica

Mica content in pounds per gallon = A X B X C

where A = percentage of mica in pigment, by weight
B = percentage of pigment in paint, by weight
C = weight per gallon paint, lb.

Less than 0.30 pound of mica per gallon of paint shall constitute failure of this test.

4.3.4.2 Spectrographic method. Use Method E-2 SM10-19, Determination of potassium in titanium dioxide: Methods of Emission Spectrochemical Analysis. The factor to convert to percent mica is: 8.45 X percent K_2O. Less than 0.30 pound of mica per gallon of paint shall constitute failure of this test.

Pages 6 and 7

4.3.9 through 4.3.9.8, delete and substitute:

4.3.9 Fungus resistance. The fungus resistance of the paint shall be determined in accordance with FED. TEST METHOD STD. No. 141, method 6241 except that:

(1) Aspergillus oryzae, ATCC 10195 or QM 1273 shall be used as the test organism,

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:14Z
Check the source to verify that this is the current version before use.

(2) The test specimens shall be leached before innoculation,
(3) The incubation period shall be 7 days, and
(4) The sterile distilled water used in the preparation of the innoculum shall contain approximately 0.005 percent of a nontoxic wetting agent.

Page 7

After 4.3.9.8, add following new paragraphs:

4.3.10 Lead. Test for lead in accordance with ASTM D2088, except that calculation of lead metal shall be as follows:

Lead (percent by weight of total solids) = A X 0.86623 X B X C X 100

where A = grams of lead oxide (PbO_2) in ash
B = percent ash by weight of paint divided by 100
C = percent total solids by weight of paint divided by 100

Presence of lead metal in excess of 0.5 percent by weight of total paint solids shall constitute failure of this test.

4.3.11 Mercury. Test for mercury in accordance with ASTM D2206. Presence of any quantity of mercury shall constitute failure of this test.

5.1.1, third subparagraph, lines 2 and 3, delete "30 percent passing a No. 70 sieve" and substitute "20 to 40 percent passing a No. 70 sieve".

Page 8

6.5, delete and substitute:

6.5 Test organism. Cultures of the organism used in the fungus resistance test specified in 4.3.9 may be obtained from the American Type Culture Collection, 12301 Parklawn Drive, Rockville, MD 20852, or for service use, The Mycology Group, Pioneering Research Laboratory, U. S. Army Natick Laboratories, Natick, MA 01760.

MILITARY INTEREST

Custodians:

Army - ME
Navy - YD

Review activities:

Army - MR, MU, EL

Preparing activity:

Army - ME

Civil Agency Coordinating Activity:

GSA-FSS

☆ U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/4082

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:14Z
Check the source to verify that this is the current version before use.